Week 2: Foundations of Market Microstructure

From the limit order book to the role of the market maker

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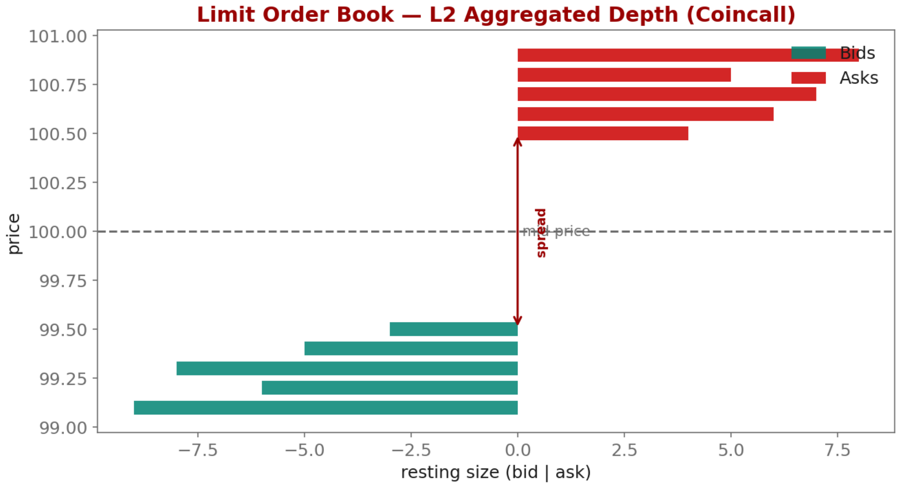
Learning Objectives

- Reconstruct an L2 order book from Coincall data
- Decompose the bid-ask spread (Roll, Glosten-Milgrom, Kyle)
- Explain the market maker as a continuous liquidity provider

Lecture 1: Order Book Mechanics

- Order types: market, limit, IOC, post-only, hidden
- Price-time priority and the matching algorithm
- L1 and L2 views (Coincall provides L2; there is no L3 / per-order feed)
- Because data is L2, we model aggregate intensity, not queue position

- Adverse selection + inventory + order-processing costs
- Roll (1984): effective spread from trade prices
- Glosten-Milgrom (1985): asymmetric information sets the spread
- Realized spread, effective spread, price impact


Limit Order Book — L2 Aggregated Depth (Coincall)
101.00
100.75
100.50
100.25
100.00
99.75
99.50
99.25
99.00
price
Bids
Asks
spread
mid-price
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
resting size (bid | ask)

Getting the Data from Coincall

- Goal: pull a single option's order book and recent trades, build mid/spread.
- Option order book (100 depth): GET /open/option/order/orderbook/v1/{symbol}
- Recent trades: GET /open/option/trade/lasttrade/v1/{symbol}
- List symbols first: GET /open/option/getInstruments/BTC
- Response gives arrays of [price, size] for bids and asks -> aggregate (L2)
- For replay/backtests use the provided Parquet L2 snapshots (1s for options)
- Live updates: subscribe to the /options/orderbook WebSocket channel

Cross-check exact paths at https://docs.coincall.com/

Code — week2_orderbook.py

```python
from coincall_client import signed_get # from Week 1

def l2_top_of_book(symbol, key, secret):
    ob = signed_get(f"/open/option/order/orderbook/v1/{symbol}", {}, key, secret)
    data = ob["data"] # confirm field names in the docs
    bids = sorted(data["bids"], key=lambda x: -float(x[0]))
    asks = sorted(data["asks"], key=lambda x: float(x[0]))
    best_bid, best_ask = float(bids[0][0]), float(asks[0][0])
    mid = 0.5 * (best_bid + best_ask)
    spread = best_ask - best_bid
    return mid, spread, best_bid, best_ask

# For an offline backtest, replay the provided Parquet instead:
# import pandas as pd
# book = pd.read_parquet("coincall_btc_options_l2_2025Q4.parquet")
# book["mid"] = 0.5 * (book.bid_px_0 + book.ask_px_0)
```

Glossary — Key Terms

Limit order book (LOB) all resting limit orders organized by price level

L1 / L2 / L3 top-of-book / aggregated depth / per-order; Coincall provides L2 only

Mid-price the average of the best bid and best ask

Adverse selection loss from trading against better-informed counterparties

Effective spread actual cost relative to mid (estimated by the Roll measure)

Price impact how far the mid moves in the direction of an incoming trade

Order-flow toxicity the degree to which incoming flow is informed

Reading

- Cartea-Jaikumar-Penalva, Ch. 1-3; Roll (1984); Glosten-Milgrom (1985)

References

- Roll, R. (1984). A simple implicit measure of the effective bid-ask spread. Journal of Finance, 39(4), 1127-1139.
- Glosten, L. R., & Milgrom, P. R. (1985). Bid, ask and transaction prices in a specialist market. J. Financial Economics, 14(1), 71-100.
- Kyle, A. S. (1985). Continuous auctions and insider trading. Econometrica, 53(6), 1315-1335.
- O'Hara, M. (1995). Market Microstructure Theory. Blackwell.
- Cartea, Jaikumar, & Penalva (2015). Algorithmic and High-Frequency Trading, Ch. 1-3.